

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

Computer Vision Engineer with 3+ years of experience delivering production computer vision applications using Python, PyTorch, TensorFlow, and OpenCV. Proven ability to design and conduct experiments to improve robustness, accuracy, and real-time performance. Experienced in application development, end-to-end ownership of CV systems, and technical leadership across edge and cloud deployments. Holds an MSc in Robotics, with a strong foundation in perception, optimisation, and experiment-driven engineering from the University of Bristol.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed and deployed production computer vision applications for real-time video analytics on edge devices delivering deterministic 24/7 operation in safety-critical oil-rig environments.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed and conducted adversarial-condition training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Designed and validated deterministic system behaviour through on-site field testing, performing root-cause analysis of model failures and reducing crash frequency by 67% in live video pipelines.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across deployments.
- Owned end-to-end data pipelines, including automated data collection, augmentation, and labelling, increasing dataset diversity by 40% and reducing manual labelling effort by 60%.
- Communicated experimental results and system performance to engineers and clients, translating quantitative metrics into actionable optimisation and deployment decisions.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led technical delivery of real-time computer vision applications for object detection, pose estimation, and tracking, achieving sub-50 ms inference latency and reducing false positives by 33% on embedded hardware.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Owned the full machine learning pipeline from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation, accelerating iteration cycles and improving deployment reliability.
- Provided technical leadership to a 4-member engineering team, setting experimentation standards, reviewing model performance, and guiding optimisation strategies for production deployments.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full application using Docker to ensure reproducibility, version control, and consistent deployment across development environments.

SKILLS

- **Perception & SLAM:** 3D Feature Extraction, Convolutional Neural Networks, Data Augmentation, Eigen, LiDAR SLAM, NumPy, OpenCV, Point Cloud Library, ROS2 image_transport, Scikit-Image, YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, Docker, DVC (Data Version Control), Git, Intel VPUs, Linux, Microsoft Azure, MLFlow
- **Programming & Development:** Bash, C++17, CMake, colcon, Django, Pandas, PostgreSQL, Python, Qt, rclcpp, rclpy, ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership